

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	Mean acceleration = 0.75[3] [m s ⁻²] (1) Absolute uncertainty = $\frac{(0.79 - 0.71)}{2} = \pm 0.04$ [m s ⁻²] (1)		2		2	2	2
		(ii)	5% (accept 5.3%) (ecf) no s.f. penalty		1		1	1	1
	(b)	(i)	$\frac{0.6}{60}$ or $\frac{0.1}{10}$ so % uncertainty = 1 [%]		1		1	1	1
		(ii)	Total mass = $\frac{60 \times 10^{-3} \times 9.81}{0.75[3]}$ ecf = 0.78 [kg] (1) Mass of glider = 0.78(ecf) – 0.06 = 0.72 [kg] (1) Total % uncertainty = 6.3% (accept 6%) (ecf from (a)(ii) and (b)(i)) (1) Absolute uncertainty in mass of glider = ± 0.045 or ± 0.05 or 0.04 if 6% used (ignore sig figs and ecf if total mass used - expect ± 0.047 (6%) or 0.049 (6.3%)) (1) Mass of glider = 0.72 ecf (even if 0.78 used) ± 0.04 or 0.05 [kg] seen (1) Sig figs need to be consistent and to a max of 2 sig figs for the uncertainty Accept: 0.722 - 0.725 ± 0.043 - 0.050 [kg] (NOT 0.72 ± 0.050 kg) 720 ± 40 [g] 720 ± 50 [g] 722 - 725 ± 43 - 50 [g]		5		5	5	5
	(c)		Reasonable suggestion e.g. release mechanism - no detail required or equivalent Move light gate to the right Increase mass <u>of glider</u> or decrease slotted mass. Don't accept increase mass only. Measure acceleration between two light gates Accept more repeat readings			1	1		1
			Question 3 total	0	9	1	10	9	10